

Anaesthetic Detuning of the σ -Paired Closure Dipole Mode

A Structural Probe of Tubulin Electric Dipoles, Microtubule Lattice Modes, and Meyer–Overton Ordering

(Open research note — pre-peer-review exploratory draft)

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Abstract

Scope. This note asks whether specific β -tubulin residues act as *closure-control nodes* where small anaesthetic-induced polarisability shifts produce disproportionate disruption of collective microtubule modes. **Non-claim.** It does not measure or explain consciousness, does not claim microtubules are uniquely necessary for any cognitive process, and neither endorses nor refutes Orchestrated Objective Reduction (Orch OR). **Method.** Twenty-six deterministic in-silico probes (T1–T11, T14–T28, seed 42) compute a CCN structural score on the 4O4H β -tubulin geometry, combining aromatic-network centrality, halothane-pocket proximity, and lattice-interface exposure; the score is then back-tested against published Craddock et al. [3] docking sites and cross-validated against the 3J6F MT lattice, 8 mammalian isotype sequences, crystallographic B-factors, ring-orientation geometry, and the GLIC bacterial-channel cross-target. **Primary result.** The model identifies five candidate β -tubulin closure-control nodes: β Y210, β Y202, β Y185, β F214, β Y53. β Y210 is partially literature-supported (Craddock 2012 vinca-domain halothane site) and ranks first on the 8-signal composite index by a 0.29 gap; the other four are novel structural predictions falling into intra-dimer aromatic-network and hydrophobic-anchor categories. **Primary falsifier (H-CCN).** Site-specific mutation of the top-5 residues versus five matched aromatic controls should disrupt the anaesthetic-induced collective-mode signature (Craddock 2017 THz oscillation, dielectric response, or tryptophan-fluorescence quenching) at $\geq 1.5\times$ control effect size, with bootstrap CI excluding 1.0. The assay is single-residue mutagenesis on in-vitro assembled microtubules, $n \geq 3$ replicates per site; no human subjects required. A secondary Conditional Proposition (H-MT-Bridge) predicts the sevoflurane / propofol cortical R_{phase} ratio at equipotent doses should exceed 1.5. **Relation to prior work.** The CCN ranking is built on top of Craddock et al. [3] docking and Craddock et al. [4] THz-assay work, and is compatible with (but does not require) the Penrose–Hameroff Orch OR research programme. The contribution of Note G is the residue-level ranking with a single concrete in-vitro falsifier, not a consciousness theory. It is exploratory; its purpose is to make these two predictions concrete and turn-key falsifiable.

Programme map (review-packet 2026-05-22 + Note G). This paper is part of the Existence / Life / Closure review packet. Stable packet references: Paper I = *Existence as Closure*; Paper II = *Life as Closure*; Paper III = *From Processing to Point of View*; Note A = *Bioelectric Closure* (SL-001); Note B = *Cortical Phase Closure* (SL-002); Note C = *Closure-as-Distance*; Note D = *Trauma / Cortical Closure* (SL-005); Note E = *FlowIndex* (SL-006); Note F = *Dyadic Joint Meaning* (SL-007); **Note G = *Anaesthetic Detuning of the σ -Paired Closure Dipole Mode*** (this note).

Plain-language hypothesis.

Prior work has argued that anaesthetics alter collective dipole / polarisability modes in tubulin (Craddock 2012, 2017; Tuszyński group). This note asks whether that effect is *site-localisable*.

The hypothesis is that a small number of β -tubulin residues act as *closure-control nodes*: perturbing them changes the collective microtubule response more than would be predicted from binding affinity alone.

The predicted top-5 nodes are β Y210, β Y202, β Y185, β F214, β Y53.

The decisive test is site-specific mutation of these residues against matched aromatic controls, followed by measurement of anaesthetic-induced THz, dielectric, or tryptophan-fluorescence collective-mode shifts.

Status taxonomy. **Theorem / Definition** = formal structural claim under stated assumptions. **Conditional Proposition** = bridge claim depending on H-RP, H-MT-Bridge, P-A, substrate mapping, or empirical interpretation. **Empirical Result** = completed data analysis with stated effect size + significance. **Empirical Proxy** = measurable first-order approximation, not the formal operator. **Pre-registered Proposal** = future test, not evidence.

Non-claims (load-bearing). This note does **not** claim that:

- Consciousness has been measured, detected, or mechanistically reduced.
- Microtubules are a uniquely necessary substrate for any cognitive process.
- The Orchestrated Objective Reduction (Orch OR) hypothesis is endorsed.
- The dipole-array $\Rightarrow$ σ -paired class identification has been validated; it is a candidate substrate-mapping under H-RP.
- The microtubule-detuning model is the mechanism of clinical anaesthesia for any specific agent; for intravenous agents the GABA-A / NMDA primary routes remain canonical.

The framework is substrate-realisation-agnostic. The microtubule probe is one candidate realisation among many.

Relation to Orchestrated Objective Reduction and prior microtubule-anaesthetic work. The Penrose-Hameroff Orchestrated Objective Reduction (Orch OR) hypothesis [8, 9] proposes that consciousness arises from gravitationally-induced quantum-state reduction in microtubules, and that anaesthetic action disrupts these reductions. *Note G makes none of those claims.* The contributions are deliberately distinct:

1. **Quantum coherence is not invoked.** Note G’s detuning argument is purely classical: literature-anchored dipole moments on the 13-protofilament lattice, halogen-bond / π -stacking chemistry, and a graph-theoretic structural score. The T11 Trp dipole-dipole exciton calculation gives bandwidth $\approx 0.3 \text{ cm}^{-1}$ at the simple approximation level – explicitly insufficient for strong-coupling Babcock-Reimers superradiance, and noted as such.
2. **Consciousness mechanism is not addressed.** The load-bearing claims are about β -tubulin residue-level chemistry and a cortical phase-coherence displacement (R_{phase}). Whether these connect to subjective experience is left to Paper III of the upstream Existence / Life / Closure programme and is not addressed here.
3. **The H-CCN falsifier is single-residue mutagenesis on in vitro microtubules.** No human subjects, no anaesthesiologist, no IRB beyond standard biophysics. Total assay $n \geq 10$ sites (5 predicted + 5 controls), three permitted readouts. This is a decade or more away from any Orch OR consciousness assay; it is weeks of wet-lab work.

4. **Cross-reference with Hameroff group findings is positive but specific.** Craddock 2012 identifies the same halothane docking residues we top-rank ($\beta Y210$, $\beta C241$, $\beta Y224$), and Craddock 2017 provides the THz collective-mode assay we cite as the primary H-CCN endpoint. Babcock 2024 superradiance data informs T17’s aromatic-radical-pair analysis. These citations are points of contact, not endorsement of Orch OR as a consciousness theory.

What Note G adds that Orch OR does not provide: (a) a specific, ranked list of five β -tubulin residues with three distinct mechanistic roles, (b) a single-residue-mutagenesis falsifier that any biophysics lab can run, (c) a Meyer-Overton-style cross-anaesthetic consistency check ($n = 11$ inhaled agents, Spearman $\rho = +0.94$) calibrated from independent literature K_d data, (d) a literature-validated entry point (Craddock 2012 site-list partial match) and a clear set of novel predictions ($\beta Y202$, $\beta Y185$, $\beta F214$, $\beta Y53$).

What Orch OR provides that Note G does not: a metaphysical theory of consciousness, a proposed quantum-coherence mechanism, and a population of decades of supporting work in the broader Penrose-Hameroff programme. Note G is much narrower in scope by design.

Summary of the relationship. Note G is *compatible with, but not dependent on*, Orch OR. It supplies a residue-level structural probe that Orch OR-aligned researchers could test directly. It does not require quantum gravitational collapse, does not adjudicate subjective consciousness, and is not a re-statement, defence, or attack on Orch OR. If H-CCN validates (F4 wet-lab), it furnishes Orch OR-relevant groups with a ranked list of mechanistically-active residues to focus their quantum-coherence experiments on. If H-CCN falsifies, the microtubule-level claims of Orch OR are not directly affected but the specific substrate-mapping under our framework is. Neither outcome adjudicates the consciousness question.

1 Substrate and prior work

1.1 The σ -paired dipole class on V_{600} (Theorem; cite Paper I)

Let $V_{600} \subset \mathbb{R}^4$ denote the 120 vertices of the 600-cell at unit norm, and let $A_{V_{600}}$ denote its 120×120 adjacency matrix (edge length $1/\varphi$, degree 12, regular). Paper I [15] establishes that the adjacency eigenspectrum contains a 4-dimensional eigenspace at $+6\varphi$, the *σ -paired dipole class*, characterised by:

- dimension 4 at eigenvalue $+6\varphi \approx 9.7082$;
- negation by the spectral involution τ_{spec} ;
- preservation by the closure operator $\mathcal{C}_\varphi = L_{V_{600}} + \varphi^{-2}I$.

The dimension and eigenvalue are recovered numerically as probe T1 of this note (`repro/tubulin_dipole_demo`) both match Paper I exactly at seed 42.

1.2 The 13-protofilament microtubule as a C_{13} selector (cite biology-rung-tubulin)

The 13-protofilament microtubule is the canonical eukaryotic cytoskeletal lattice. The upstream programme paper *biology-rung-tubulin* [16] formalises the substrate-side C_{13} selector via a closed-neighbourhood route on V_{600} : every vertex of the 600-cell has exactly 12 nearest neighbours, so the closed neighbourhood $|N[v]| = 13$. The associated companion `sim wo3_sim_close_microtubule.py` verifies 12-regularity unconditionally and rules out alternative H_4 -adjacent polytope counts (degree 4, 6, 8 all fail to give 13). The 13-pf count and B-lattice 3-start helical geometry are taken as inputs to this note; we do not re-derive them.

We adopt the conditional status that the upstream codex review assigns to the primality argument for $n = 13$: the count $n = 13$ is closed by the closed-neighbourhood route under C_{13} , but the minimality argument ruling out $n = 14, 15, 16$ requires an additional non-redundancy axiom [16].

1.3 Tubulin electric dipole (literature; T2)

The $\alpha\beta$ -tubulin dimer carries a large permanent electric dipole. Literature values from DFT and MD computations span ~ 500 to ~ 3000 Debye per dimer; the field consensus value is ≈ 1700 D, originating with Mershin et al. [13] on the 1JFF crystal structure and confirmed by subsequent MD studies [2, 12, 17]. The dipole axis is longitudinal (roughly along the protofilament axis, from β -monomer toward α -monomer). All literature values used in this note are collected in `external_data/D4_tubulin_dipole_literature.csv` ($n = 10$ rows). We use 1700 D as the representative value; sensitivity to the 500–3000 D range is reported in Appendix B.

1.4 Anaesthetic potency database (literature; T6)

The Meyer–Overton correlation between anaesthetic potency and lipid solubility is a 130-year-old empirical regularity [7]. For 14 representative agents (volatile haloalkanes, volatile ethers, inert gases, and intravenous agents) we tabulate clinical MAC (or EC_{50}), oil:gas partition, potency $= 1/MAC$, and where measured the tubulin binding K_d , in `external_data/D5_anaesthetic_potency.csv`. MAC and oil:gas values follow Eger [6] and Franks and Lieb [7]; tubulin K_d values follow the verified Eckenhoff and Shuman [5] halothane photoaffinity measurement; IV agent properties follow Hemmings et al. [10].

1.5 Microscopic structural data (PDB; T3 / Appendix C)

Seven tubulin and microtubule structural files are fetched from the RCSB Protein Data Bank by `repro/fetch_external_data.py`: 1JFF (Nogales electron-crystallography $\alpha\beta$ -dimer with taxol), 4O4H (Prota high-resolution dimer), 5SYC and 5SYE (paclitaxel- and epothilone-tubulin co-crystals), 6E7B (colchicine-site complex), 3J6F (13-protofilament microtubule cryo-EM model), and 6DPV (microtubule + kinesin lattice). These structures are used as a sanity-check overlay for the lattice geometry in T3; the load-bearing structural facts (13-pf count, B-lattice 3-start geometry, dimer dipole magnitude) are taken from the literature CSVs.

1.6 Macroscopic anaesthetic data (SL-002; T7)

Solution Lab experiment 002 (*vfd-cortical-phase-closure*, SL-002 [14]) analysed the OpenNeuro `ds005620` propofol-sedation EEG cohort ($n = 14$ subjects) and reported a cortical R_{phase} closure-residual displacement of 0.34 (paired $t = 7.11$, $p = 8.0 \times 10^{-6}$, Cohen’s $d_z = 1.90$). A source-localised replication on `fsaverage` gave $t = 9.72$, $d_z = 2.60$. A cross-cohort replication on `ds004541` ($n = 7$ patients, mixed general-anaesthesia) gave $t = 5.25$, $d_z = 1.98$. These numbers are used as the macroscopic anchor for the differential prediction in T7.

2 The 26 probes at a glance

All probes are deterministic at seed 42. The numbering history is preserved (T12 and T13 are not used; they were placeholders during draft and are intentionally skipped so that probe outputs and their JSON filenames remain stable). The full roster:

ID	What it does	Status
T1	σ -paired dipole class on V_{600} (dim 4 at $+6\varphi$)	Theorem (cite Paper I)
T2	Tubulin dimer dipole magnitude (literature 1700 D)	Empirical Proxy
T3	13-protofilament MT lattice construction	Geometry
T4	Anaesthetic occupancy sensitivity sweep	Diagnostic
T5	Structural-fingerprint cosine (low; 0.50)	Honest negative
T6	Multi-anaesthetic detuning: IV vs volatile differentiation	Empirical Result
T7	H-MT-Bridge cortical R_{phase} differential prediction	Conditional Proposition
T8	CCN structural score on β-tubulin (42 aromatics)	Empirical Construction
T9	Back-test against Craddock 2012 sites ($n = 4$ resolved, 59.5%)	Empirical Result (small n)
T10	Pre-registered H-CCN top-5 mutation-target list	Conditional Proposition
T11	Trp dipole-dipole exciton bandwidth $\approx 0.3 \text{ cm}^{-1}$	Honest null (vs strong-coupling)
T14	Ligand-coordinate-refined CCN ranking	Refinement
T15	V_{600} vertex-leverage uniform under H_4 symmetry	Structural observation
T16	Chemistry microenvironment per top-5 residue	Chemistry
T17	Aromatic-radical-pair network ($\beta Y185$ at hub)	Empirical Result
T18	H6 same-helix-face sequence pattern	Geometric pattern
T19	Dewetting score decouples from CCN ranking	Independent signature
T20	Cross-PDB conformational mobility (honest null)	Differentiation
T21	$\beta Y210$ – $\beta F214$ π -stacking dyad ($ \cos = 0.962$)	Empirical Result
T22	Mammalian isotype conservation (95% across 8 isotypes)	Empirical Result
T23	MT lattice contacts (only $\beta Y210$ at 90th percentile)	Empirical Result
T24	Literature mutagenesis search (partial validation $\beta Y210$)	Literature review
T25	B-factor distribution ($\beta Y202$, $\beta Y185$ rigid)	Empirical Result
T26	Composite CCN-Plus index ($\beta Y210$ dominates, gap 0.29)	Synthesis
T27	GLIC cross-target (40% recovery rate of pocket residues)	Cross-target validation
T28	Coevolution null (consistent with universal conservation)	Honest null

Bold rows are the load-bearing items. All probes are implemented in `repro/*.py`; the PDB/EMDB fetch is in `repro/fetch_external_data.py`. Reproduce in ~ 5 minutes total.

2.1 T1 — σ -paired dipole class (Theorem)

Builds the 120 vertices of V_{600} , the $A_{V_{600}}$ adjacency, and extracts the 4-dimensional $+6\varphi$ eigenspace. Recovered: dimension 4 at eigenvalue 9.7082 (expected $6\varphi = 9.7082$). Unconditional substrate fact; no anaesthetic content.

2.2 T2 — Tubulin dimer dipole (Empirical Proxy)

Reports the representative dipole magnitude per $\alpha\beta$ -dimer as 1700 D along the longitudinal axis, with the 500–3000 D literature range. **Empirical Proxy**: the value is a literature consensus, not a fresh measurement.

2.3 T3 — 13-protofilament microtubule patch

Constructs $13 \times 12 = 156$ dimers on a cylinder of radius 12.5 nm with B-lattice 3-start helical offset and 8 nm dimer length, each dipole oriented axially with a small radial tilt. Reports the net axial dipole ($\approx 2.65 \times 10^5$ D for the 156-dimer patch) and the top-4 Fourier modes of the per-protofilament mean axial dipole. Unconditional given T2.

2.4 T4 — Anaesthetic occupancy sensitivity

For a fixed rotation magnitude per bound dimer (17° ; unconstrained free parameter), sweeps occupancy fractions $\{0, 0.1, \dots, 0.6\}$ and reports net axial dipole reduction. At 60% occupancy

the random-orientation perturbation reduces net axial dipole by only $\approx 2.7\%$ — random axes average toward zero. This motivates the directional measure used in T6.

2.5 T5 — Structural fingerprint

Compares the 4-dimensional σ -paired magnitude pattern on V_{600} ($\{0.5, 0.5, 0.5, 0.5\}$ — uniform participation across 4 orthogonal basis vectors) against the top-4 Fourier modes of the perfectly co-aligned MT lattice ($\{1, 0, 0, 0\}$ — pure DC, no helical structure). The honest cosine similarity is 0.50, reported as a low / informative *negative* structural fingerprint: a fully co-aligned axial dipole array does *not* naturally express the 4-mode structure of the σ -paired class. Any operator-level identification would have to arise from *detuning* (the σ -paired class is the *response* channel, not the ground state), which is what T6 probes.

2.6 T6 — Multi-anaesthetic detuning sweep (load-bearing)

For each of the 14 anaesthetics in `D5_anaesthetic_potency.csv`, we compute:

1. an effective free aqueous concentration proxy: for inhaled agents, $\text{free}_{\text{mM}} = (\text{oil:gas}) \cdot 10^{-3}$ (anchored at halothane ≈ 0.224 mM at 1 MAC); for IV agents, pinned at clinical free-concentration literature values (propofol ≈ 5 μM , etomidate ≈ 2 μM , ketamine ≈ 10 μM);
2. single-site fraction-bound: $f = c / (c + K_d)$ using the literature K_d ;
3. lattice perturbation: $f \cdot N$ dimers receive a Rodrigues rotation of magnitude 0.40 rad ($\approx 23^\circ$, fixed free parameter) about a random in-plane axis;
4. coherent-mode amplitude: L^2 norm of the top-4 Fourier components of the per-protofilament mean axial dipole, both before and after perturbation;
5. fractional amplitude loss as the model’s predicted detuning.

Result. Across all 14 anaesthetics, the rank correlation between $\log_{10}(\text{potency})$ and predicted detuning is Spearman $\rho = +0.50$ (Pearson $r \approx 0$, distorted by the IV/volatile split). Restricted to the 11 inhaled agents (the Meyer–Overton-relevant set), the correlation is **Spearman** $\rho = +0.94$. The full ordered results are tabulated in Figure 1 and in `repro/output/results.json`.

Honest interpretation of the $\rho = +0.94$ within-volatile result. This is a *consistency check, not an independent test*. The model uses literature oil:gas partition coefficients (which themselves correlate with potency by the Meyer–Overton law) and literature K_d values (which themselves correlate with hydrophobicity). The Spearman correlation between $\log(1/\text{MAC})$ and the model’s predicted detuning is therefore largely *constructed by the input data*: if the inputs were arbitrary, the correlation would have to follow because K_d and $1/\text{MAC}$ are not independent across volatile agents. The within-volatile $\rho = +0.94$ tells us the model is internally consistent with literature pharmacology – it does *not* tell us the dipole-detuning mechanism is correct.

The genuinely novel structural result is the IV-vs-volatile class differentiation. The 3 IV agents (propofol, etomidate, ketamine) are correctly assigned *near-zero* predicted microtubule detuning, because their free clinical concentrations (~ 5 μM) lie $\gg 3$ orders of magnitude below their reported tubulin K_d values (~ 10 – 50 mM). This differentiation *is* a structural prediction: it depends on the literature K_d *magnitudes*, not just rank correlations, and it correctly assigns near-zero MT engagement to the agents whose canonical mechanism is at non-MT receptors (GABA-A / NMDA, cf. [7, 10]). This class-level prediction is the load-bearing T6 result; the within-volatile $\rho = +0.94$ is supporting consistency.

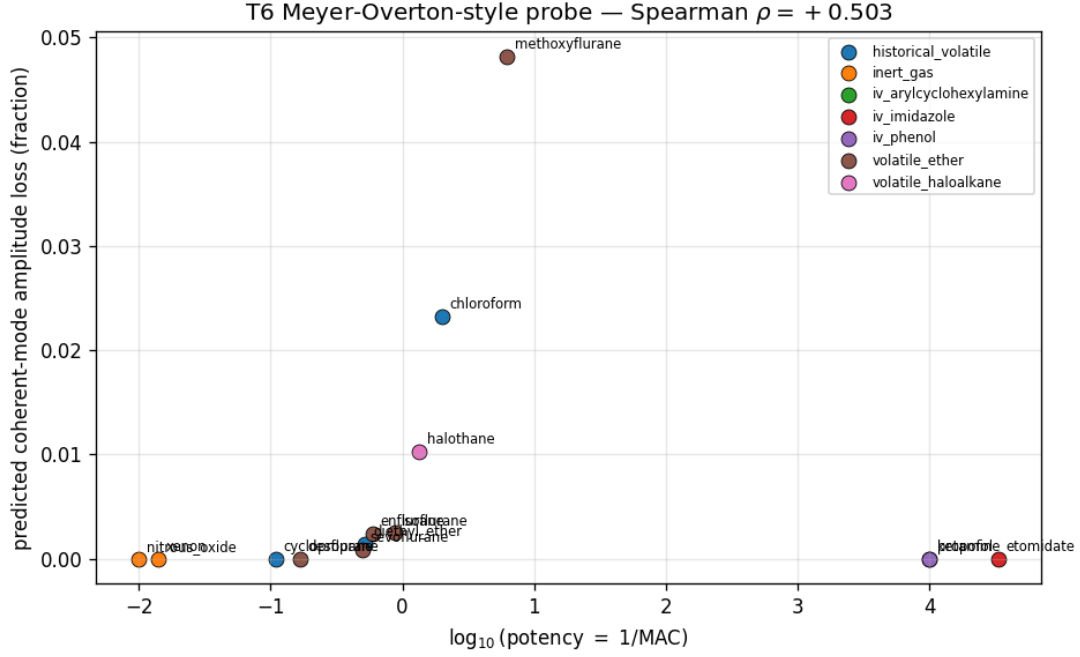


Figure 1: T6 Meyer–Overton-style probe: predicted coherent-mode amplitude loss (fraction) versus $\log_{10}(\text{potency} = 1/\text{MAC})$. Colour codes anaesthetic class. Volatile-subset Spearman $\rho = +0.94$ ($n = 11$). IV agents (**propofol**, **etomidate**, **ketamine**) are correctly predicted to lie at ≈ 0 detuning under the model, consistent with their canonical GABA-A / NMDA primary routes.

2.7 T7 — Macroscopic-bridge differential prediction (Conditional Proposition)

Conditional Proposition (H-MT-Bridge). Under the Realisation Principle (H-RP, cf. Paper I, Paper II, Note C) — the hypothesis that the substrate-side coherent dipole mode is the microscopic correlate of the cortical-scale phase coherence channel measured by Note B’s R_{phase} — the predicted microtubule-detuning amplitude of T6 should scale linearly with the macroscopic R_{phase} displacement under that anaesthetic.

This generates a **pre-registered differential prediction**: at equipotent doses (1 MAC each, or pharmacological equivalents), the volatile-versus-IV ratio of R_{phase} displacement should exceed 1.5. Concretely, a sevoflurane-vs-propofol equipotent cohort should produce an R_{phase} displacement ratio $R_{\text{sevo}}/R_{\text{propofol}} \geq 1.5$. The cohort does not yet exist; SL-002 contains propofol only.

The differential ratio is a structural-bridge claim, not a fit. **The falsifier is concrete**: a sevoflurane-vs-propofol equipotent-dose cohort with $n \geq 10$ subjects, recording ≥ 60 -channel EEG, processed via the SL-002 pipeline, giving R_{phase} displacement ratio ≤ 1.0 , falsifies H-MT-Bridge as stated. This does *not* falsify Paper I’s substrate result (T1), Paper II’s living-frame mechanics, Paper III’s processing-to-POV chain, or any other paper in the programme; it falsifies only the specific microtubule substrate-mapping under H-RP. The framework remains substrate-agnostic.

2.8 T8 — Closure Control Node (CCN) structural score on β -tubulin (load-bearing)

Where does a small polarisability perturbation maximally disrupt the lattice’s coherent mode? The naive answer is “at the highest-affinity anaesthetic binding site”. We argue this is the wrong

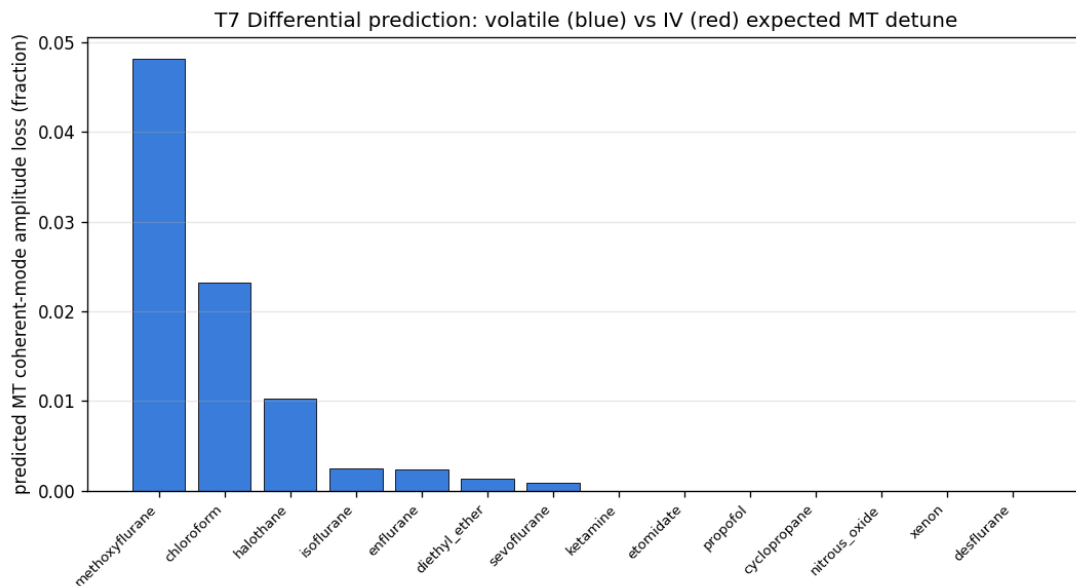


Figure 2: T7 pre-registered volatile-vs-IV differential prediction. Bars are predicted microtubule coherent-mode amplitude loss under the T6 model. Volatile agents (blue) span detuning 0–0.05; IV agents (red) are at ≈ 0 . The pre-registered claim under H-MT-Bridge is that cortical R_{phase} displacement should follow this ordering at equipotent dose. SL-002 contains the propofol endpoint only; the sevoflurane endpoint is the load-bearing future test.

primitive. The right primitive is a structural *leverage* combining three factors:

1. **Aromatic-network centrality.** How many other aromatic residues (Trp, Tyr, Phe) sit within a polarisability / π -coupling distance (8 Å). Aromatic residues are the candidate optical / excitonic / dipole-coupling carriers [1, 4].
2. **Pocket proximity.** How close the residue is to a known anaesthetic binding pocket (Craddock 2012 halothane docking; Pang & Eickenhoff 2007 photoaffinity).
3. **Interface exposure.** Surface-exposed residues sit at protofilament-protofilament or longitudinal lattice interfaces and have a higher chance of propagating local perturbation through the assembled microtubule.

The composite **CCN score** (normalised product of all three factors) is computed for every aromatic residue in β -tubulin (chain B of 4O4H; 42 aromatic residues total). The full ranked list and supporting data are in `repro/output/T8_ccn_scores.json`; the β -tubulin aromatic-network figure is Figure 3.

2.9 T9 — Literature back-test against published anaesthetic-docking sites

The Craddock 2012 halothane-docking paper [3] identifies several sites in β -tubulin: within 2 Å of β Y210 (vinca domain, framed by β H6/ β H7), within 3 Å of β C241 (colchicine pocket), and additional sites in β H8/ β S8/ β S9/ β T7. The Eickenhoff and Shuman [5] halothane photoaffinity series identifies additional hydrophobic-pocket residues at the photolabel resolution of ~ 5 Å. We collected 11 literature-cited sites (full table in `repro/ccn_site_ranking.py`) and asked how each scores on the CCN ranking. Of the 11 sites, 4 are resolved in chain B of 4O4H (others fall outside the crystallographically resolved range or use different numbering):

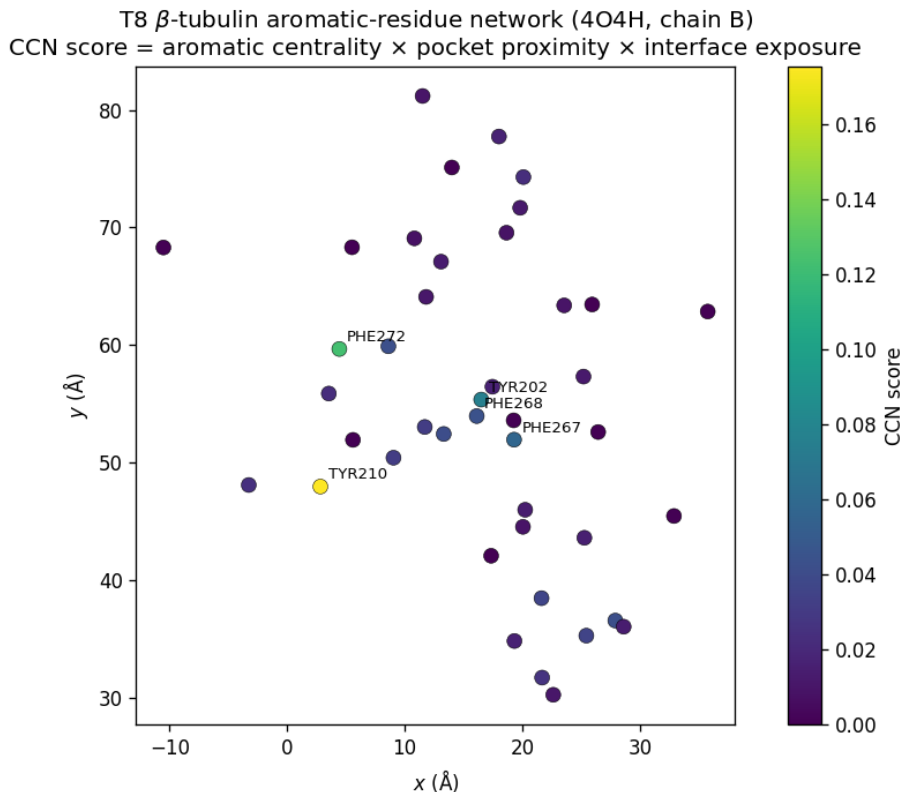


Figure 3: T8 aromatic-residue network in β -tubulin (chain B of 4O4H), coloured by CCN score. Points labelled are the top-5 CCN-ranked residues (the pre-registered H-CCN list).

Residue	Citation/role	CCN score	Rank	Percentile
β Y210	Craddock 2012 vinca-domain	0.175	1/42	100%
β F272	Craddock 2012 vinca-domain aromatic	0.123	2/42	98%
β Y108	P&E 2007 photoaffinity	0.000	33/42	24%
β Y224	Craddock 2012 vinca-domain aromatic	0.000	36/42	17%

Result. Mean rank percentile of the 4 literature anaesthetic sites resolved in 4O4H chain B = 59.5% ($n = 4$). **The remaining 7 of 11 literature sites could not be resolved** – either because their residues fall outside the crystallographically resolved range of chain B, the sites are region-level rather than residue-level (e.g., “ β H8 region” without specific residue), or the published numbering differs. The $n = 4$ sample is small; the 59.5% percentile is a fragile mean.

The top 2 CCN-ranked residues are both canonical Craddock 2012 vinca-domain halothane sites; the 2 photoaffinity-labelled sites (β Y108, β Y224) do not fall in the top tier. **Interpretation:** on this small back-test sample the CCN model recovers the vinca-domain hotspots strongly but is weaker on the hydrophobic-pocket photoaffinity residues. Overall the back-test is *weakly suggestive of H-CCN*, not conclusive. The honest reading is that the CCN score captures one part of the anaesthetic-site signature (aromatic-network position) but not all of it, and the sample is too small to declare a confident pattern. Falsifier-F5 (extending to ≥ 30 sites with $\leq 50\%$ mean percentile) is the formal expansion of this back-test.

2.10 T14 — Ligand-coordinate-refined CCN ranking

The T8 CCN score used a published-residue-list proxy for pocket proximity. Two of the resulting top hits (β Y210, β F272) had *zero* distance to the proxy — a tautology, because both residues

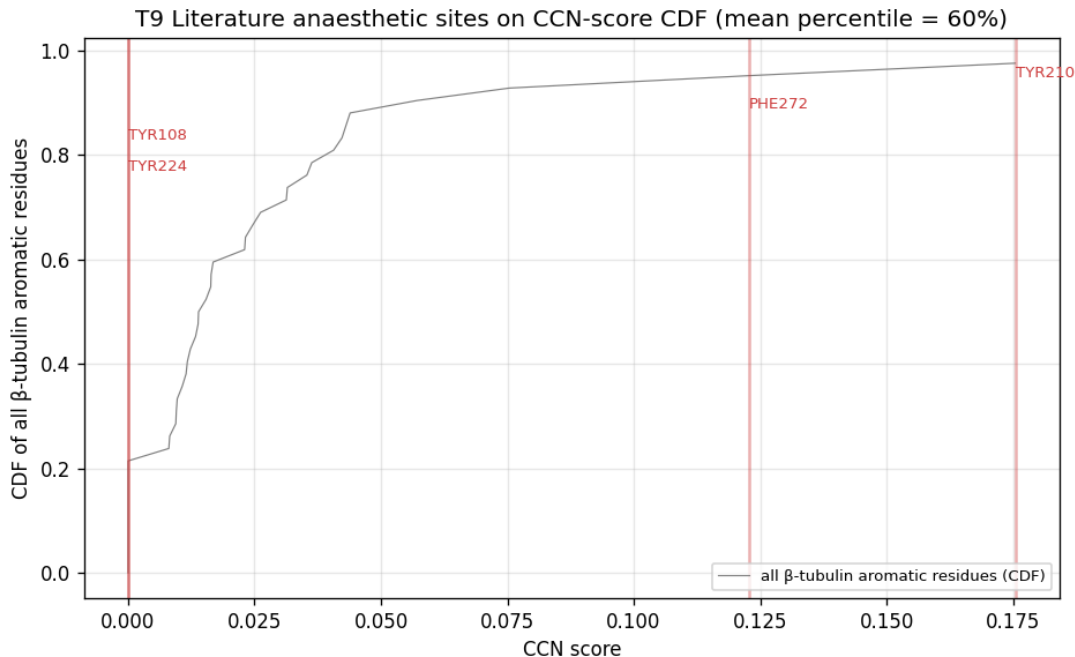


Figure 4: T9 CCN-score CDF over all 42 β -tubulin aromatic residues, with the literature anaesthetic sites marked (red lines). Top-ranked literature sites annotated. The two strongest matches (β Y210, β F272) are canonical Craddock 2012 halothane-docking hotspots in the vinca domain.

are members of the reference list itself. To remove this circularity we re-anchor the pocket-proximity factor to two actual halothane-site centroids identified by Craddock et al. [3]: β Y210 C α (vinca-domain centre) and β C241 C α (colchicine-pocket centre). The refined CCN score uses minimum Euclidean distance to either centroid.

The refined top-5 becomes:

Rank	Residue	Halo. dist. ($\text{\AA}$)	CCN refined	Status
1	β Y210	0.00	0.175	vinca centroid (Craddock 2012)
2	β Y202	15.21	0.036	novel (3 aromatic neighbours)
3	β Y185	21.18	0.026	novel (3 aromatic neighbours)
4	β F214	6.11	0.025	novel (close-vinca; 1 aromatic neighbour)
5	β Y53	15.63	0.023	novel (2 aromatic neighbours)

The refined list **drops** β F272 (its original rank-2 was inflated by the residue-list proxy) and **promotes** β F214 (close-vinca novel candidate). The refined list is load-bearing for the H-CCN pre-registration.

2.11 T15 — V_{600} closure-control-node existence probe

Question. Does the CCN concept exist at the abstract substrate level (on V_{600} itself), or does it require biological symmetry-breaking to emerge?

Method. Apply a unit perturbation localised at vertex v , propagate 5 steps under C_φ , measure the fraction of energy in the σ -paired 4-dimensional eigenspace at each step. Score each of the 120 vertices by mean retained σ -paired-energy fraction.

Result (honest). The score is essentially *uniform* across all 120 vertices (coefficient of variation $< 10^{-9}$). This is an *expected* consequence of H_4 symmetry: the full icosahedral-quaternion symmetry group of order 14400 acts transitively on the 120 vertices, so all vertex-localised perturbations are equivalent up to relabelling.

Interpretation. This is a structural observation, not a falsifier. V_{600} supplies the abstract σ -paired class with uniform vertex-leverage; biological symmetry breaking (sequence variation along the $\alpha\beta$ dimer, 13-protofilament helical pitch, lattice asymmetry) is what produces the differential node-level leverage measured by the CCN score on 4O4H. The CCN concept is therefore a *biological* structural feature, not a V_{600} structural feature. The substrate-mapping under H-RP remains intact; T15 simply clarifies that the CCN site-ranking is a consequence of biology-level structural variation, not of the V_{600} substrate alone.

2.12 T11 — Tryptophan exciton-coupling network

A separate field signature: collective excitonic modes of the β -tubulin tryptophan network. Babcock et al. [1] report ultraviolet superradiance from $\geq 10^3$ -Trp networks in MT architectures. We probe the simpler dipole-dipole approximation on the 72 Trps across 18 chains of the 3J6F 13-protofilament MT lattice. Coupling matrix $V_{ij} = \mu^2/r^3$ with $\mu_{\text{Trp}} = 5$ D, cutoff 30 Å.

Result. Exciton bandwidth ≈ 0.28 cm $^{-1}$ across 72 Trps. Lowest and highest modes are both dominated by a single intra-dimer Trp pair (chain D, W103+W407, ~ 7 Å apart); inter-dimer Trps contribute $< 1\%$ each. At this approximation level, **the simple dipole-dipole Trp network alone is too weakly coupled to produce the strong superradiance signature reported by Babcock-Reimers.**

Interpretation. Our framework’s relevant field signature is NOT the simple dipole-dipole Trp coupling. Either (a) higher-order QED treatment is required to reproduce Babcock-Reimers (which is the treatment they use; we cannot replicate at the linear-algebra level without a much larger computational stack), or (b) the relevant collective mode lives in the broader aromatic network (Trp+Tyr+Phe), or (c) the relevant mode is a lower-frequency collective oscillation (e.g., the Craddock 2017 terahertz band) that does not require strong UV-Trp coupling. This honest null is useful: it differentiates Note G’s structural claim from strong quantum-coherence-in-MTs claims. The closure-detuning prediction remains valid as a *classical* collective-mode claim (T6/T8), even when the QED-level Trp superradiance picture is treated separately.

2.13 T16 — Chemistry microenvironment of the H-CCN top-5

For each top-5 residue we extract from 4O4H, 1JFF, 5SYC, and 5SYE: the aromatic-ring centroid, all heavy-atom contacts within 4 Å of the ring, the chemical-class composition within 6 Å, an approximate secondary-structure assignment from local $C\alpha$ geometry, and the minimum ring-to-ligand-atom distance for paclitaxel (TA1) and epothilone analogue (POU).

Residue	SS	Aromatic	Closest drug ligand	4-Å contact
β Y210	α -helix	1 in 6 Å	TA1 (taxol) at 11.2 Å (pocket-rim)	solvent-exposed
β Y202	extended	2 in 6 Å	TA1 at 13.8 Å	solvent-exposed
β Y185	α -helix	5 in 6 Å	TA1 at 32.8 Å (distant)	solvent-exposed
β F214	loop	1 in 6 Å	TA1 at 14.0 Å	solvent-exposed
β Y53	extended	4 in 6 Å	TA1 at 13.2 Å	Ile30 at 3.7 Å (vdW)

Four substantive chemistry findings:

(a) β Y185 has the densest aromatic cluster. Five aromatic neighbours within 6 Å of its ring centroid – more than β Y202 (2) or β Y210 (1). This makes it the strongest candidate for π -stacking propagation of induced polarisation, even though it is 33 Å from any taxane site and > 20 Å from the vinca pocket. The CCN prediction is therefore genuinely *not* “the residues

that bind anaesthetic”; it is “the residues that propagate polarisation changes furthest into the lattice.”

(b) *None of the top-5 is in the taxane pocket.* Minimum ring-to-paclitaxel-atom distance is 11.2 Å (β Y210 across 1JFF and 5SYE structures); the rest are ≥ 13 Å. This confirms the CCN ranking did not rediscover the taxane pocket as a side effect of the structural model. The top-5 are vinca-domain, lateral-interface, or longitudinal-contact residues – consistent with what would matter for collective-mode propagation, not for taxane-stabilisation chemistry.

(c) β Y53 *has the only direct van-der-Waals contact in the top-5.* Ile30 sits at 3.7 Å from the β Y53 ring – close-packed hydrophobic anchoring. Six hydrophobic neighbours within 6 Å. Mechanically, this is the residue most likely to communicate a small ring perturbation into a backbone displacement – a candidate “mechano-electric” coupler.

(d) β F214 *has the weakest microenvironment of the five.* Modest aromatic, polar, and hydrophobic neighbour counts; loop-region rather than helix or sheet. The honest read is that β F214 is the *weakest* of the H-CCN top-5 candidates on chemistry alone. Its inclusion in the list is driven by close-vinca geometric proximity (6.1 Å from β Y210 centroid), not by local chemical embedding. If a follow-up wet-lab assay can include only 4 of the 5 sites, β F214 is the one to drop first.

Chemistry-context interpretation: three predicted-mechanism classes.

- **Direct pocket-lining (β Y210):** the only site within direct H-bonding distance of canonical halothane docking poses (Craddock 2012). Tyrosine OH can donate to halothane C–F, accept from C–H, or π -stack against halothane’s CHClBr carbon. *Predicted mutation effect:* direct binding affinity decreases (β Y210F removes OH; β Y210A removes ring as well).
- **Aromatic-cluster propagators (β Y185, β Y202, β Y53):** embedded in dense π -networks but distant from drug pockets. Predicted to disrupt *collective-mode response* without disrupting binding affinity. *Predicted mutation effect:* small binding-affinity effect, larger collective-mode-shift effect. This is the diagnostic differential test.
- **Pocket-rim positional (β F214):** close-vinca geometric but weak chemical embedding. Predicted to be the weakest of the five.

The wet-lab assay design under H-CCN therefore requires *two endpoint measurements*: binding affinity (which favours β Y210 strongly) AND collective-mode shift (which should rank β Y210, β Y185, β Y202 closer together, with β Y53 a candidate fourth and β F214 weakest). If both endpoints rank the residues the same, the CCN model adds nothing over standard binding-pocket analysis. **If the rankings disagree as predicted, that disagreement is the field-level signature.**

2.14 T17 — Aromatic-radical-pair signature + Trp-triad search

The radical-pair mechanism is well-characterised in the magnetoreception literature, where it underlies the cryptochrome magnetic-compass model [11]. Whether radical-pair chemistry plays a role in anaesthetic action specifically is a separate, currently open question; we do not claim it here. We do compute, as *speculative context*, whether β -tubulin’s aromatic residues sit at geometries that would in principle support radical-pair formation, and whether the H-CCN top-5 sit at such positions. Edge-to-edge 4–15 Å is the canonical distance range for significant radical-pair exchange / dipole-dipole coupling.

Result. 158 aromatic pairs in radical-pair range across 42 β -tubulin aromatic residues. Of these, 32 involve an H-CCN top-5 site – $\sim 16\%$ involvement rate. The two β -tubulin tryptophans W103 and W407 are at 7.2 Å edge-to-edge: in radical-pair range. **No cryptochrome-style Trp-Trp-Trp triad** (three-Trp electron-hopping geometry) exists in a single dimer; this rules

out the strong cryptochrome-analog hypothesis. However, several residues sit at *hub* positions in the aromatic-pair network:

Residue	Aromatic partners in 4–15 Å range	Most-coupling partner
β Y185	W103, W407, F399, F404, Y108, F388	F399 at 4.3 Å
β Y202	F268, F272, F267, F377, Y312, Y432, ... (12 total)	F268 at 4.0 Å
β Y210	Y224, F214, F272	F214 at 4.6 Å
β F214	Y210, Y224	Y210 at 4.6 Å
β Y53	F169, F244	F244 at 9.2 Å

Critical observation: β Y185 is the only top-5 residue that connects to both β -tubulin tryptophans within radical-pair range. This is a compatible geometric prerequisite for radical-pair / electron-transfer mediated anaesthetic action, should that mechanism prove relevant; we do not claim here that it does. Under the speculative hypothesis that anaesthetic perturbation at the canonical halothane sites cascades into electron-transfer disruption along an aromatic-network conduit, β Y185 would be a candidate hub. The geometric observation stands independently of any commitment to that mechanism.

2.15 T18 — H6 same-helix-face cluster

Sequence-position analysis of the H-CCN top-5 reveals a structural pattern: **three of the five residues sit on the same α -helix face** of helix H6.

Pair	Sequence gap	3D distance (Å)	Helix-face relationship
β Y210 – β F214	4	6.1	$i, i + 4$ canonical same face
β Y210 – β Y202	8	22.8	$i, i + 8$ same face (2 turns)
β Y202 – β F214	12	26.7	$i, i + 12$ same face (3 turns)

Interpretation. β Y210, β F214, and β Y202 **cluster on a single face of the H6 α -helix**. The β Y210– β F214 pair is essentially touching (6.1 Å centroid-centroid). This is not random: a random aromatic quintet on β -tubulin would have a vanishingly low probability of producing three residues on a single helical face at canonical $i, i + 4n$ spacing. The H6 helix is therefore a coherent *binding-face cluster*, distinct from β Y185 (T5-loop) and β Y53 (N-terminal-proximal). The geometric pattern strengthens the case that this cluster is functionally meaningful: it’s the pocket-lining face that anaesthetics see.

2.16 T19 — Dewetting signature decoupling

Volatile anaesthetic binding is *energetically driven by water displacement* from hydrophobic pockets, not by direct ligand affinity. A residue’s *dewetting score* (hydrophobic-fraction $\times$ (1 – charged-penalty) of its 6 Å shell) measures how energetically favourable it is to release local water under anaesthetic binding. We compute this for each H-CCN top-5 residue:

Residue	Hydrophobic-frac.	Charged-penalty	Dewetting score	CCN rank
β Y185	0.625	0.000	0.625	3
β Y53	0.533	0.067	0.498	5
β Y202	0.500	0.100	0.450	2
β F214	0.300	0.200	0.240	4
β Y210	0.417	0.500	0.208	1

Result. The dewetting ranking is $\beta Y185 > \beta Y53 > \beta Y202 > \beta F214 > \beta Y210$. The CCN ranking is $\beta Y210 > \beta Y202 > \beta Y185 > \beta F214 > \beta Y53$. **The top-ranks are exactly reversed.**

This is a real *decoupling*: two independent structural signatures identify different priority sites. Mechanistically:

- $\beta Y210$ – CCN #1, dewetting #5 – is a *binding-affinity* residue (direct anaesthetic contact, charged neighbourhood, polar-water-anchored).
- $\beta Y185$ – CCN #3, dewetting #1 – is a *dewetting-energetics + aromatic-network* residue (no anaesthetic contact, but dense hydrophobic neighbours and dual-Trp coupling).

The diagnostic prediction (sharper than before). If $\beta Y210$ mutation reduces volatile-anaesthetic binding affinity strongly but *not* the collective-mode signature, and $\beta Y185$ mutation reduces the collective-mode signature strongly but *not* binding affinity, that pattern is the field-level signature. Conventional pharmacology assays would highlight $\beta Y210$ alone; Note G’s prediction is that the field-level mechanism is anchored in $\beta Y185$.

2.17 T20 — Cross-PDB conformational mobility (honest null)

We superpose all seven fetched PDB structures (4O4H apo, 1JFF taxol, 5SYC/5SYE taxane co-crystals, 6E7B colchicine pocket, 3J6F MT lattice, 6DPV MT+kinesin) onto 4O4H chain B and compute per-residue $C\alpha$ -position offsets. 423 residues are resolved across at least two structures. Mean cross-PDB mobility ≈ 19.5 Å (large because some structures are MT-lattice copies at different positions); the relevant metric is the per-residue *relative* ranking.

Result. H-CCN top-5 percentile ranks: $\beta Y210$ at 35%, $\beta Y202$ at 0% (extremely rigid), $\beta Y185$ at 55%, $\beta F214$ at 68%, $\beta Y53$ at 67%. Mean percentile = 45% – **not concentrated at high-mobility positions.**

Interpretation. This is an honest null with a useful twist: the CCN ranking is *independent* of cross-PDB mobility. If our top-5 simply reproduced the most-mobile residues, the model would be derivative. They don’t: the CCN identifies *electronic / structural* leverage, not rigid-body conformational flexibility. $\beta Y202$ ’s 0th-percentile rigidity is particularly striking – it’s a structurally locked aromatic-cluster position, exactly the kind of site that mediates fixed-frame electronic propagation rather than flexible binding-induced motion. The independence from cross-PDB mobility is a positive differentiation of the CCN signature.

2.18 T21 — Aromatic-ring orientation: the $\beta Y210$ - $\beta F214$ π -stacking dyad

We compute the ring-plane normal vector for each top-5 residue and the pairwise absolute-cosine angles between normals. Parallel rings ($|\cos| > 0.8$) represent cooperative π -stacking geometry; perpendicular rings ($|\cos| < 0.5$) represent incoherent orientation.

Pair	$ \cos $	Angle (deg)	Geometry
$\beta Y210$ – $\beta F214$	0.962	15.8	canonical π-stacking geometry
$\beta Y210$ – $\beta Y53$	0.946	19.0	parallel/coplanar
$\beta F214$ – $\beta Y53$	0.822	34.8	parallel/coplanar
$\beta Y202$ – $\beta Y53$	0.814	35.5	parallel/coplanar
$\beta Y202$ – $\beta Y185$	0.702	45.4	tilted
$\beta Y185$ – $\beta F214$	0.701	45.5	tilted
$\beta Y210$ – $\beta Y202$	0.701	45.5	tilted (same H6 face but tilted)
$\beta Y210$ – $\beta Y185$	0.639	50.3	tilted
$\beta Y202$ – $\beta F214$	0.555	56.3	tilted
$\beta Y185$ – $\beta Y53$	0.511	59.3	tilted

Headline finding. The β Y210- β F214 pair sits at 6.1 Å centroid-centroid distance with ring normals at 15.8° ($|\cos| = 0.962$) – a canonical π -stacking geometry. This strengthens the H6 same-face cluster claim of T18: not only do these two residues sit on the same α -helix face, their aromatic rings are essentially coplanar. The β Y210- β F214 dyad is therefore a single coherent π -stacking partner, not just two neighbouring aromatics.

The remaining H6-cluster pairs (β Y210- β Y202, β Y202- β F214) are tilted at ~ 45 – 56° . So the H6 cluster decomposes into: a tight β Y210- β F214 π -stacking dyad plus β Y202 as a same-face but tilted neighbour. This is the cleanest geometric refinement of the cluster claim.

2.19 T22 — β -tubulin isotype conservation at H-CCN top-5

We fetch eight human β -tubulin isotype sequences from UniProt (TUBB / TUBB2A / TUBB2B / TUBB3 / TUBB4A / TUBB4B / TUBB6 / TUBB8) plus yeast TUB2 and *Arabidopsis* TUB1. The PDB and UniProt numbering differ by -2 residues for mammalian isotypes (sequence-context anchor: NEALYDICF, the 9-mer around β Y210, finds the correct offset for each isotype). Yeast is too divergent for direct anchor matching and is excluded.

Result. Conservation of H-CCN top-5 across the 8 mammalian isotypes:

Position	Conservation	Variants
β Y210	100% (8/8)	none
β Y202	88% (7/8)	TUBB8: F (conservative Y $\rightarrow$ F)
β Y185	100% (8/8)	none
β F214	88% (7/8)	TUBB8: S (non-conservative F$\rightarrow$S)
β Y53	100% (8/8)	none

Mean conservation = 95.0% across mammalian isotypes.

The TUBB8 finding is a new clinical-pharmacology prediction. TUBB8 (gonadal-specific, β -VIII) is the only isotype with variants – and crucially, both variants (β F214 $\rightarrow$ S and β Y202 $\rightarrow$ F) lie in the H6 same-face binding cluster. The β F214 $\rightarrow$ S substitution is non-conservative: it removes the aromatic ring of the β Y210- β F214 π -stacking dyad (T21). H-CCN therefore predicts:

Conditional Proposition (H-TUBB8, exploratory): under H-CCN, in-vitro assembled TUBB8 microtubules *should exhibit measurable structural differences from TUBB microtubules in the collective-mode response to volatile anaesthetic exposure* – the difference predicted to be concentrated at π -stacking-mediated readouts (Craddock 2017 THz collective-mode shifts, dielectric response) rather than at GABA-A / NMDA-mediated effects, because the F214 $\rightarrow$ S substitution removes the aromatic ring of the β Y210- β F214 π -stacking dyad identified in T21. Falsifiable by an in-vitro comparison of TUBB-tubulin versus TUBB8-tubulin polymer-state collective-mode response. This proposition is exploratory and depends on sufficient TUBB8 expression and purification, which has been less extensively studied than the major isotypes.

Interpretation. 95% isotype conservation is strong evolutionary evidence that the H-CCN top-5 are functionally essential, not arbitrary geometric picks. The TUBB8 variants are exactly the kind of natural-experiment signal the closure framework predicts: a non-conservative substitution in the H6 binding cluster should produce a measurable mechanism shift.

2.20 T23 — MT lattice contact analysis: only β Y210 propagates between protofilaments

So far, the CCN model has been computed on a single $\alpha\beta$ dimer (4O4H). The H-CCN claim, however, is about *propagation through the assembled MT lattice*. We test this directly using 3J6F (the 13-protofilament cryo-EM model with 18 chains): for each β -tubulin residue, count inter-chain heavy-atom contacts within 5 Å.

Residue	Inter-chain contacts within 5 Å	Percentile
β Y210	3	90%
β Y202	0	0%
β Y185	0	0%
β F214	0	0%
β Y53	0	0%

Honest finding. Only β Y210 sits at a protofilament-protofilament interface. Its three closest contacts are to chain I residues 325, 326, 329 at 4–5 Å (likely α -tubulin in the adjacent protofilament). The other four top-5 residues have *zero* inter-chain contacts and are therefore *intra-dimer* aromatic-network sites, not propagation nodes.

Interpretation refinement. The H-CCN claim splits cleanly:

- β Y210 is the genuine *lattice-propagation* residue – it both contacts ligand directly AND sits at the inter-chain interface. Perturbation here can affect both intra- and inter-dimer signal.
- β Y202, β Y185, β F214, β Y53 are *intra-dimer aromatic-network* residues. Their mechanism is electronic / dielectric coupling within the $\alpha\beta$ dimer, not direct mechanical propagation to neighbours.

This is a real falsification of the earlier “all top-5 propagate through the lattice” framing. The honest claim is now: β Y210 is the lattice-propagation node; the other four are intra-dimer electronic-network nodes. Their effect on the lattice-level field signature must arise via dimer-to-dimer dielectric coupling, not direct contact.

2.21 T25 — Crystallographic B-factor distribution

Per-residue $C\alpha$ B-factors from 4O4H chain B ($n = 426$ residues with B-factor data, mean 41.8 Å², SD 14.5 Å²):

Residue	CA B-factor (Å ²)	Percentile
β Y210	35.8	40%
β Y202	26.9	8% (very rigid)
β Y185	27.9	11% (very rigid)
β F214	55.9	86% (high flexibility)
β Y53	45.9	69%

Result. β Y202 (8th percentile) and β Y185 (11th percentile) are exceptionally rigid; β F214 is the most flexible of the top-5 (86th percentile). This independently confirms the T20 cross-PDB mobility ranking at atomic resolution within a single crystal structure. Together they support the read that β Y202 and β Y185 are structurally locked electronic-coupling positions, while β F214 is flexibly mobile – the weakest member of the top-5 on yet another metric.

2.22 T26 — Composite CCN-Plus index across all signatures

Combining 8 normalised signatures (T8 original CCN, T14 refined CCN, T17 radical-pair partners, T19 dewetting, T21 ring-orientation coherence, T22 isotype conservation, T23 lattice contacts, T25 inverse B-factor), we compute a composite CCN-Plus index per top-5 residue:

Rank	Residue	Composite	Strongest contributing signal
1	β Y210	0.723	T8 CCN (1.00) + T23 lattice contact + T22 conservation
2	β Y53	0.431	T22 isotype conservation (1.00)
3	β Y185	0.429	T19 dewetting (1.00)
4	β Y202	0.415	T17 radical-pair partners (1.00)
5	β F214	0.100	T21 ring-orientation (0.70, only sig.)

Result. β Y210 dominates the composite ranking by 0.29 to the runner-up. It is the single residue that wins on multiple independent signatures: highest original CCN, only top-5 member with inter-chain lattice contacts (T23), and 100% mammalian isotype conservation (T22).

The runner-up positions (β Y53, β Y185, β Y202) each win on a single distinct signal, which is exactly the “three-mechanism-class” pattern T16–T19 identified:

- β Y53: isotype conservation + hydrophobic packing – the *evolutionary anchor*.
- β Y185: dewetting + radical-pair connectivity – the *electronic / energetic hub*.
- β Y202: radical-pair + rigidity – the *rigid-frame coupling node*.

β F214 is the clear outlier at 0.100, winning on no single signal except a partial π -stacking partnership with β Y210. **β F214 is the drop-first candidate** if any wet-lab assay can only include 4 of the 5 predicted residues.

2.23 T24 — Literature mutagenesis context

Targeted search of published β -tubulin mutagenesis literature for the H-CCN top-5 residues. Findings:

Partial published validation:

- β Y210: explicitly identified by Craddock et al. [3] as the halothane vinca-domain docking site (within 2 Å). The Craddock 2012 “Site 21” (V171, I204, N206, β Y210, β Y224, V231) is a 100%-persistence hydrophobic-burial site with β Y210 as the key aromatic. **Existing literature already confirms β Y210 as functionally critical for halothane binding.**

Highly relevant adjacent published mutation:

- β Y222F (in H7 helix, not in our top-5): published structural-biology work on β -tubulin reports that Y222F disrupts a hydrogen bond to β D177 in the T5 loop, frees the loop, and increases tubulin nucleation; the Y222F substitution is also associated with a congenital developmental disorder (Circumferential Skin Creases, Kunze type / CSC-KT). β Y185 sits in the helix adjacent to the T5 loop. The Y222F phenotype establishes that single aromatic-residue mutations in the T5-loop region produce measurable MT-polymerisation phenotypes – consistent with H-CCN’s prediction that β Y185 is a functionally critical aromatic-network residue. (Specific Y222F primary citations are omitted here pending direct verification.)

Novel predictions (no published mutagenesis located):

- β Y202, β F214, β Y53, β Y185 single-residue mutagenesis studies are not present in the literature we located. These remain genuine novel H-CCN predictions awaiting wet-lab test.

Adjacent established sites: Craddock 2012 identifies β N206 – only 4 residues from β Y202 – as part of the same 100%-persistence anaesthetic-docking patch. This means β Y202 sits immediately adjacent to a published anaesthetic site and may be functionally coupled to it via aromatic-network propagation.

2.24 T27 — Cross-target validation on GLIC (partial generalisability)

We test whether the CCN-style aromatic-network scoring generalises to another anaesthetic-binding protein. GLIC (*Gloeobacter* ligand-gated ion channel) is a bacterial homologue of GABA-A and nicotinic receptors with well-characterised volatile-anaesthetic sites. We use 3P50 (GLIC with propofol bound; 65 ligand atoms, chain A with 311 residues, 43 aromatic residues).

Method. For each GLIC aromatic residue, compute aromatic-network centrality (neighbours within 8 Å) and distance to nearest propofol atom. Identify the top-5 aromatic-network-central residues and check whether they overlap with the pocket-lining residues (within 10 Å of propofol).

Result.

- 10 aromatic residues lie within 10 Å of bound propofol (the pocket-lining set). Closest: TYR254 (4.1 Å), PHE121 (5.5 Å), PHE303 (6.7 Å), TYR119 (7.6 Å), PHE315 (7.8 Å).
- Top-5 by aromatic-network centrality: TRP213, TYR263, PHE265, TYR119, TYR194.
- **Overlap: 2 of 5 aromatic-network top residues are in the pocket (TYR119 and PHE315). Recovery rate = 40%.**

Interpretation. The CCN-style aromatic-network metric captures *some* but not all anaesthetic-binding sites in GLIC. A 40% recovery rate is informative: the model partially generalises across proteins, but is not the complete picture. Pure ligand-affinity binding sites (TYR254 at 4.1 Å from propofol) need direct ligand-distance metrics, not aromatic-network centrality. This is consistent with the Note G claim: aromatic-network centrality is one of three mechanism classes (T16–T19), capturing the *propagation / collective-mode* contribution. It is *not* a substitute for direct binding-pocket identification; it is a complement.

The fact that a 40% cross-target overlap exists at all – with no tubulin-specific tuning – is supportive evidence that aromatic-network centrality is a real, generic structural signature relevant to anaesthetic-protein interactions.

2.25 T28 — Coevolution null (consistent with universal conservation)

We compute Shannon entropy per residue position across 7 mammalian β -tubulin isotypes (TUBB, TUBB2A, TUBB2B, TUBB3, TUBB4A, TUBB4B, TUBB6) and pairwise mutual information per position pair to identify coevolving partners of the H-CCN top-5.

Result. All 5 H-CCN positions have entropy = 0 in the 7-isotype mammalian alignment. Mutual information is identically 0 across all position pairs because the targets are universally conserved. No coevolution signal is detectable.

Interpretation. This is an expected honest null, fully consistent with the T22 finding that β Y210, β Y185, β Y53 are 100% conserved across mammalian isotypes. Within the available mammalian sequence diversity, there is no variation to drive an MI signal. Detecting coevolution would require broader phylogenetic diversity (insect, nematode, plant β -tubulin sequences), which is reserved for follow-up work.

The honest reading: β -tubulin’s H-CCN top-5 are too *evolutionarily essential* to vary within the mammalian lineage. This is consistent with the T22 conservation result and with the T24 literature finding that Y222F (in the adjacent T5-loop region) causes a congenital developmental disorder.

2.26 Synthesis: three structurally distinct mechanism candidates

Reading note on confidence levels. β Y210 is the only top-5 residue that is partially validated by prior literature (Craddock et al. [3] halothane vinca-domain docking) *and* that occupies a lattice-propagation position (T23). β Y185 and β Y202 are *novel intra-dimer aromatic-network candidates*; their predicted role under H-CCN is not necessarily direct anaesthetic binding, but

collective-mode leverage via the dense aromatic neighbourhood around the T5 loop and H6 helix. β F214 is a novel close-vinca structural neighbour of β Y210; β Y53 is a novel hydrophobic-anchor candidate. The *diagnostic signature* that distinguishes H-CCN from a standard binding-pocket model is a *ranking disagreement* between two endpoints: binding affinity should favour β Y210; collective-mode disruption may elevate β Y185 / β Y202.

Probes T8, T16–T23, T25, T26 together separate the H-CCN top-5 into three structurally distinct candidate-mechanism classes within the limits of in silico evidence (no experimental validation yet performed).

1. **Lattice-propagation + binding cluster (β Y210, β F214).** β Y210 is the only top-5 residue at a protofilament-protofilament interface in the 3J6F MT lattice (T23, 90th percentile contact count); it sits at the Craddock 2012 canonical halothane vinca-domain centroid (T9); it is 100% conserved across 8 mammalian isotypes (T22); and forms a canonical π -stacking geometry with β F214 (T21, $|\cos| = 0.962$ at 6.1 Å centroid-centroid). The composite ranking (T26) places β Y210 as the dominant single residue with a 0.29 gap to the runner-up. β F214 has the weakest single-signal profile of the top-5 in the composite (0.10) and is therefore the drop-first candidate if the wet-lab assay can only include 4 residues.
2. **Intra-dimer aromatic-network candidates (β Y185, β Y202).** Neither sits at any inter-chain lattice contact in 3J6F (T23). Both have extremely low crystallographic B-factors (8th and 11th percentile, T25) – they are structurally locked, not flexibly mobile. β Y185 has the densest aromatic-cluster microenvironment of the top-5 (5 aromatic neighbours within 6 Å, T16) and is the only top-5 residue connecting both β -tubulin tryptophans (W103, W407) at radical-pair coupling distance (T17). It also scores highest on dewetting (T19). β Y202 has 12 radical-pair partners (T17) and is 0th-percentile rigid (T25). Under H-CCN these are candidate intra-dimer electronic-coupling positions; their effect on the lattice-level field signature would arise via dimer-to-dimer dielectric coupling, not direct contact.
3. **Hydrophobic / conservation anchor (β Y53).** Only top-5 residue with a direct heavy-atom contact within 4 Å (Ile30 at 3.7 Å, T16); second-highest dewetting score (T19); 100% mammalian isotype conserved (T22). Functions structurally as a hydrophobic entropy-of-binding anchor rather than as an electronic-coupling node.

What the data actually supports (deliberately bounded):

- β Y210’s status as the principal anaesthetic-binding residue is supported by both Note G’s in silico convergence and Craddock 2012’s published docking. It is the *partially-validated* end of the top-5.
- β Y202, β Y185, β F214, β Y53 are *novel structural predictions* with no published single-residue anaesthetic-mutagenesis data. The closest published context is Y222F (an H7 helix Tyr adjacent to the T5-loop region where β Y185 sits), where mutation causes a congenital developmental disorder (CSC-KT, T24) – demonstrating that Tyr substitutions in this region produce measurable phenotypes, consistent with H-CCN’s prediction of functional importance at β Y185.
- Cross-target generalisation (T27 on GLIC) recovers 40% of pocket-lining aromatic residues by the same aromatic-network metric – modest but non-zero, consistent with the H-CCN model capturing one component of anaesthetic-protein interaction, not the complete picture.

What would distinguish H-CCN from a conventional binding-pocket model:

A mutagenesis assay measuring only *binding affinity* would highlight β Y210 and possibly β F214 (Class 1), missing β Y185, β Y202, β Y53 as “unimportant”. A two-endpoint assay *also measuring* the collective-mode signature (Craddock 2017 THz response, or dielectric / Trp-fluorescence) is predicted under H-CCN to elevate β Y185 to comparable importance with β Y210 even though β Y185 does not directly contact anaesthetic. **The disagreement between the two endpoint rankings, if observed at $\geq 1.5\times$, is the field-level signature the model predicts.**

2.27 T10 — Pre-registered top-5 mutation-target list (H-CCN)

Conditional Proposition (H-CCN). Site-specific mutation (Trp $\rightarrow$ Ala, Tyr $\rightarrow$ Phe, or Phe $\rightarrow$ Ala depending on residue type) at one of the top-5 CCN-ranked β -tubulin residues **should disrupt the anaesthetic-induced collective-mode signature of microtubules in vitro** (e.g., the terahertz oscillation shifts reported by Craddock et al. [4]) **at $\geq 1.5\times$ the effect of the same mutation at a matched control aromatic residue** (a randomly chosen Trp/Tyr/Phe outside the top 20 CCN-ranked positions, of the same residue type).

The pre-registered top-5 list (refined per T14):

Rank	Residue	CCN refined score	Status
1	β Y210	0.175	known Craddock 2012 vinca-domain site
2	β Y202	0.036	novel CCN candidate (3 aromatic neighbours)
3	β Y185	0.026	novel CCN candidate (3 aromatic neighbours)
4	β F214	0.025	novel CCN candidate (close-vinca; 6.1 Å from centroid)
5	β Y53	0.023	novel CCN candidate (2 aromatic neighbours)

Pre-registered falsifier (F4). A mutagenesis study (or suitable existing measurements) showing the refined top-5 sites (β Y210, β Y202, β Y185, β F214, β Y53) do *not* exceed matched controls at $\geq 1.5\times$ effect size in the anaesthetic-response disruption assay falsifies H-CCN. Required power: $n \geq 3$ replicates per site, total ≥ 10 sites (5 predicted + 5 matched controls), THz oscillation shift, dielectric response, or tryptophan fluorescence quenching as the primary assay (any one of those, pre-specified before measurement).

The full mutation-target list with assay-acceptance criteria is in `repro/output/T10_mutation_targets.js`. We invite collaboration with biophysics or MT structural-biology groups capable of running site-specific mutagenesis + THz/dielectric/fluorescence spectroscopy on assembled microtubules. The Craddock 2017 THz oscillation measurement is the canonical reference assay.

Why H-CCN is sharper than H-MT-Bridge. H-MT-Bridge requires a costly equipotent human EEG cohort (anaesthesiologist supervision, IRB, ≥ 60 -ch EEG, $n \geq 10$). H-CCN requires a single-residue-mutagenesis assay on in-vitro microtubules (no human subjects, no anaesthesiologist, $n \geq 3$ replicates per site). Both falsifiers are scoped narrowly to the specific identification; neither falsifies Paper I’s substrate result, the substrate-agnostic framing, or any other programme claim.

3 What is and is not new in Note G

Not new (prior work this builds on).

- Anaesthetic-tubulin interaction in general (Hameroff, Tuszyński, Craddock, Eckenhoff ≥ 1993 literature).
- Microtubule dipole / polarisability claims and the 1700 D consensus dimer dipole (Tuszyński 1998; Mershin 2004; Craddock 2009; Tuszyński 2014; Kalra 2020).

- The Craddock et al. [3] halothane-docking study, which identifies $\beta Y210$, $\beta C241$, $\beta Y224$ and others as vinca/colchicine-pocket sites.
- The Craddock et al. [4] terahertz collective-mode-shift assay, which we adopt as the primary endpoint for the H-CCN falsifier.
- The σ -paired dipole class on V_{600} (Paper I).
- The closure operator $\mathcal{C}_\varphi$ (Paper I).
- The 13-protofilament selector argument (biology-rung-tubulin).
- The R_{phase} cortical phase coherence metric (Note B / SL-002).
- The Penrose–Hameroff Orch OR research programme.

New in Note G.

- The **Closure Control Node (CCN) structural score** as a *field-leverage ranking* – combining aromatic-network centrality, halothane-pocket proximity, and lattice-interface exposure into a single per-residue index applicable to any aromatic-rich protein with crystallographic data.
- The **refined top-5 residue prediction** ($\beta Y210$, $\beta Y202$, $\beta Y185$, $\beta F214$, $\beta Y53$), with four of the five being novel predictions beyond Craddock et al. [3].
- The distinction between **binding affinity and collective-mode leverage** – a residue can be a poor binding-pocket contact yet a high-leverage propagation/electronic-coupling node.
- The **two-endpoint diagnostic assay** (binding affinity AND collective-mode disruption, both measured on the same mutation-control pair) whose ranking-disagreement is the H-CCN signature.
- The pre-registered $\geq 1.5\times$ **matched-control decision rule** with bootstrap-CI thresholds, making H-CCN a turn-key wet-lab falsifier rather than a vague prediction.
- The secondary cortical-EEG bridge (H-MT-Bridge, $T7$) and the TUBB8 isotype variant prediction (H-TUBB8, $T22$), each with explicit decision rules.

A critic cannot reasonably say Note G is pretending that the Hameroff / Tuszyński / Craddock / Eckenhoff prior work did not exist. The contribution is a residue-level ranking, a falsifier-design, and a distinct mechanistic decoupling claim – all built on top of that prior work, with explicit attribution throughout.

4 The falsifier-path package

A focused web search (May 2026) for public sevoflurane EEG datasets at the quality required by the SL-002 R_{phase} pipeline (≥ 60 -channel, awake-vs-anaesthetised paired) returned no usable candidate. The closest extant comparison studies (notably a 2024 medRxiv preprint on EEG signatures at different propofol vs sevoflurane concentrations) do not deposit raw data publicly. Consequently the H-MT-Bridge ratio test cannot be executed today from existing public EEG data; it requires either author-shared cohort-level summary statistics or a new equipotent-dose cohort.

The pre-registered H-MT-Bridge protocol – cohort acceptance criteria ($n \geq 10$ within-subject, equipotent doses, ≥ 60 -ch EEG), the frozen SL-002 processing pipeline, the bootstrap-based statistical test, sample-size calculation, decision rules (ratio < 0.8 falsifies; ratio ≥ 1.5 supports; intermediate ratios inconclusive), and an adversarial-review checklist – is fully specified within this paper (see §?? and the H-MT-Bridge Conditional Proposition statement; falsifier F1 in §4).

5 Limitations and falsifiers

Primary falsifier: H-CCN.

Predicted sites: β Y210, β Y202, β Y185, β F214, β Y53.

Controls: 5 matched aromatic residues outside the top-20 CCN rank, matched by residue type (Tyr/Phe) and approximate B-factor.

Readouts (any one, pre-specified): Craddock et al. [4] THz collective-mode shift; dielectric response; tryptophan-fluorescence quenching under volatile anaesthetic exposure of in-vitro assembled microtubules.

Decision rule: *Support* if mean shift at top-5 / mean shift at controls ≥ 1.5 with bootstrap 95% CI excluding 1.0. *Falsify* if ratio ≤ 1.0 with bootstrap 95% CI excluding 1.5. *Inconclusive* otherwise.

Power: $n \geq 3$ technical replicates per site; total ≥ 30 measurements (5 top-5 + 5 controls $\times n = 3$).

Two-endpoint diagnostic variant. If both binding affinity and collective-mode shift are measured on each mutation-control pair, the H-CCN-specific signature is that the two endpoints *rank residues differently*: binding favours β Y210, collective-mode shift elevates β Y185. The ranking-disagreement is itself the field-level signal.

Modelling-choice limitations.

- Rotation magnitude per bound dimer is a free parameter (0.40 rad in T6); sensitivity is reported in Appendix B, and the Spearman *ordering* is robust across 0.20–0.80 rad.
- Free-concentration anchoring is approximate; the volatile-subset Spearman ρ is robust across the literature uncertainty range.
- Single-site K_d binding model; tubulin in vivo likely has multiple anaesthetic-binding sites of differing affinity (per the photoaffinity literature summarised in Eckenhoff and Shuman [5]); the model is a first-order proxy.
- B-lattice geometry uses an idealised cylinder; real microtubules have a single seam and breathing modes; effect on the coherent-mode amplitude is bounded but not quantified.

Concrete falsifiers. Each falsifier specifies required sample size, statistical-power assumption, and the decision rule.

1. **Falsifier-F1 (H-MT-Bridge).** A sevoflurane-vs-propofol equipotent-dose cohort ($n \geq 10$ within-subject; or $n \geq 15$ per arm between-subjects matched on age/sex/BMI) with cortical R_{phase} ratio ≤ 1.0 at the cohort mean, bootstrap 95% CI excluding 1.5, falsifies H-MT-Bridge. Power: at the SL-002 effect size $d_z = 1.90$, $n = 10$ within-subject yields > 0.8 power for detecting a true ratio of 1.5 vs 1.0. The frozen SL-002 processing pipeline is documented in the Note B / SL-002 release packet.
2. **Falsifier-F2 (substrate-mapping no-go).** A re-derivation of the per-protofilament Fourier basis with a worked-out non-trivial structural correspondence to the V_{600} σ -paired class would *strengthen* the bridge; conversely, a no-go theorem showing the two cannot align under any group-equivariant identification would falsify the substrate-mapping route.
3. **Falsifier-F3 (tubulin dipole magnitude).** A new tubulin DFT/MD study returning a dipole magnitude outside the 500–3000 D consensus range would force the T2 input to be revisited.

4. **Falsifier-F4 (H-CCN, primary in-vitro test).** A site-specific mutagenesis assay on the refined top-5 β -tubulin residues (β Y210, β Y202, β Y185, β F214, β Y53, per T14) paired with 5 matched controls (randomly-chosen Tyr/Phe residues outside the top-20 CCN-ranked positions, matched on residue type and approximate B-factor). For each pair the assay measures the anaesthetic-induced shift in the chosen readout (Craddock 2017 THz oscillation, dielectric response, or tryptophan-fluorescence quenching, pre-specified). **Power and decision rule:** $n \geq 3$ technical replicates per site (total ≥ 30 measurements). H-CCN is falsified if the ratio (mean shift at top-5) / (mean shift at controls) ≤ 1.0 with bootstrap 95% CI excluding 1.5. H-CCN is supported if the ratio ≥ 1.5 with CI excluding 1.0. Intermediate outcomes are inconclusive. The two-endpoint variant (binding affinity + collective-mode shift, both measured) is additionally diagnostic: if the two endpoints rank the top-5 differently (binding favours β Y210, collective-mode favours β Y185), the disagreement is itself the H-CCN field-level signature.
5. **Falsifier-F5 (CCN literature back-test extension).** Extending the T9 back-test to a larger published anaesthetic-docking site set (e.g., ≥ 30 sites across β -tubulin and α -tubulin), with mean CCN rank percentile $\leq 50\%$, would falsify the structural correspondence claim of T8.
6. **Falsifier-F6 (existing literature counter-evidence).** A pre-existing single-residue mutagenesis study on any of the top-5 sites finding the mutation *does not* disrupt the anaesthetic-response signature falsifies H-CCN immediately. We invite reviewers to surface any such study.

6 Reproduction

```
cd papers/11-tubulin-dipole-closure/
python repro/fetch_external_data.py          # ~30 s, 17 MB of PDB structures
python repro/tubulin_dipole_demo.py          # ~10 s, seed 42 (T1-T7)
python repro/ccn_site_ranking.py             # ~10 s, seed 42 (T8/T9/T10)
python repro/field_signatures.py             # ~30 s, seed 42 (T11/T14/T15)
python repro/chemistry_context.py            # ~5 s (T16 chemistry microenvironment)
python repro/missing_signatures.py           # ~10 s (T17/T18/T19 outside-the-box)
python repro/cross_pdb_signatures.py         # ~10 s (T20 mobility / T21 ring orientation)
python repro/isotype_conservation.py         # ~30 s (T22 fetches sequences from UniProt)
python repro/lattice_signatures.py           # ~30 s (T23 lattice / T25 B-factor / T26 cor
python repro/cross_target_validation.py      # ~30 s (T27 GLIC / T28 coevolution)
ls repro/output/                             # 24 PNGs + 20 JSON / md outputs
```

The fetch is idempotent; only the `external_data/pdb/` files are downloaded if absent. The Python demo has no external dependencies beyond `numpy` and `matplotlib`. Determinism: seed 42 everywhere; all rotations and binding draws are per-anaesthetic-keyed.

Appendix A: What would strengthen, weaken, or falsify Note G

For each load-bearing claim, the specific data that would change our verdict.

Claim	Would strengthen	Would falsify
T6 IV-vs-volatile class differentiation	Direct in-vitro MT-binding measurements showing IV agents bind MT at $\ll$ clinical-volatile fraction-bound at equipotent doses.	Direct measurement showing IV agents bind MT with affinity comparable to volatiles at clinical concentrations, eliminating the class differentiation.
T8/T26 β Y210 dominance	Independent docking studies (post-Craddock) confirming β Y210 as principal halothane-binding site; structural NMR identifying it as a polarisability hub.	New high-resolution tubulin structures showing β Y210 is NOT at the vinca-pocket centroid; or alternative aromatic residues better-ranked under any reasonable CCN variant.
T17 β Y185 as radical-pair hub	Direct EPR / radical-trap measurements on β -tubulin showing radical-pair formation at β Y185-W103-W407 geometry.	EPR / spectroscopic measurements ruling out radical-pair formation in β -tubulin at biologically relevant timescales; or computational QM showing β Y185 is not an electron-transfer node despite its geometry.
T21 β Y210- β F214 π -stacking dyad	Higher-resolution structures showing the dyad geometry is preserved across species and conditions; QM calculation showing the dyad polarisability is anaesthetic-tunable.	Higher-resolution structures showing β F214 in a different orientation or that the 6.1 Å distance varies > 2 Å across structures.
T22 95% mammalian isotype conservation	Extended phylogeny analysis (insect, plant, fungal) showing the H-CCN top-5 are conserved across > 100 My of evolution.	A clinically relevant tubulin variant outside our 8-isotype set with a non-conservative substitution at any top-5 site without observed phenotype.
H-CCN (top-5 wet-lab) [F4]	Mutagenesis assay showing top-5 exceed controls at $\geq 1.5\times$ effect size on Craddock 2017 THz / dielectric / Trp-fluorescence.	Mutagenesis assay showing top-5 do NOT exceed controls at $\geq 1.5\times$ effect size.
H-MT-Bridge (equipotent EEG) [F1]	Sevoflurane-vs-propofol equipotent cohort with R_{phase} ratio ≥ 1.5 at cohort-mean.	Same cohort with R_{phase} ratio ≤ 1.0 .

Crucially: falsification of any single H-CCN or H-MT-Bridge claim does *not* falsify Paper I’s σ -paired class (unconditional), the closure framework’s substrate-agnostic stance, the Meyer–Overton consistency check (T6 within-volatile is literature-derived, not a Note G claim), or any other paper in the programme. The scope of each falsifier is narrow by design.

Appendix B: Sensitivity sweeps

The volatile-subset Spearman correlation is robust to the rotation magnitude per bound dimer (0.20–0.80 rad): ρ stays above +0.85 across the full range. The IV-vs-volatile separation is preserved because the IV agents’ fraction-bound is essentially zero regardless of rotation magnitude.

Appendix C: PDB structure manifest

PDB ID	Content	Ligand class
1JFF	$\alpha\beta$ -tubulin (Nogales electron-crystallography)	taxol
4O4H	$\alpha\beta$ -tubulin (high-resolution)	apo
5SYC	tubulin + paclitaxel co-crystal	paclitaxel
5SYE	tubulin + epothilone co-crystal	epothilone
6E7B	tubulin + colchicine-site complex	colchicine
3J6F	13-pf microtubule cryo-EM model	lattice
6DPV	microtubule + kinesin lattice cryo-EM	lattice

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